Kinematic Trajectory Optimization of a Reference Clip

Objective function

Companion to `reference_kto_formulation.tex`, Sec. 3.2

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Objective

The whole-clip problem minimizes

$$J(\boldsymbol{Q}) = \sum_{t=0}^{N-1} f_t(\boldsymbol{q}_t) \;+\; w_a \sum_{t} \big\| (\boldsymbol{q}_{t+1} \ominus \boldsymbol{q}_t) - (\boldsymbol{q}_t \ominus \boldsymbol{q}_{t-1}) \big\|^2 \tag{1}$$

over the trajectory $\boldsymbol{Q} = (\boldsymbol{q}_0, \ldots, \boldsymbol{q}_{N-1})$, subject to the hard constraints (C1)–(C8) of the full formulation. The first term is the per-frame intent cost, the second the squared discrete acceleration (smoothness); $\ominus$ is the tangent-space difference (logarithm map on the base rotation, plain subtraction elsewhere).

Let

$$\mathcal{S}_t = \big\{ h : \; \sigma_{h,t} = 0, \; (h,t) \notin \mathcal{T}_{\text{rake}} \big\} \tag{2}$$

denote the pads that are released and not raking at frame t. The per-frame intent cost is

$$\begin{aligned}
f_t(\boldsymbol{q}) = \; & \sum_{h \in \mathcal{S}_t} \big\| W_h^p \big(\boldsymbol{p}_h(\boldsymbol{q}) - \boldsymbol{p}_h^*(t) \big) \big\|^2 && \text{swing-pad arc} \\
& + \sum_{h \in \mathcal{S}_t} w_h^r \big\| \boldsymbol{e}_r\big(R_h(\boldsymbol{q}), R_h^*(t) \big) \big\|^2 && \text{swing-pad orientation} \\
& + \sum_{h:\; \beta_{h,t}=1} \big\| W^{\text{toe}} \big(\boldsymbol{p}_{\text{toe},h}(\boldsymbol{q}) - \boldsymbol{p}_{\text{toe},h}^*(t) \big) \big\|^2 && \text{toe brace} \\
& + \big\| W^b \big(\boldsymbol{p}_b - \boldsymbol{p}_b^*(t) \big) \big\|^2 + w_r^b \big\| \boldsymbol{e}_r\big(R_b, R_b^*(t) \big) \big\|^2 && \text{base intent} \\
& + \big\| W_\theta \big(\boldsymbol{\theta} - \boldsymbol{\theta}^*(t) \big) \big\|^2 && \text{posture reference} \\
& + \big\| W_{\text{lim}}(\boldsymbol{\theta}) \big(\boldsymbol{\theta} - \boldsymbol{\theta}_{\text{bnd}} \big) \big\|^2 && \text{limit-margin spring} \\
& + w_\gamma^2 \|\boldsymbol{\gamma}\|^2. && \text{gimbal return}
\end{aligned} \tag{3}$$

Here $\boldsymbol{p}_h(\boldsymbol{q}), R_h(\boldsymbol{q})$ are the pad-frame position and orientation from forward kinematics, $\boldsymbol{e}_r(R, R^*) = \frac{1}{2} \sum_{k=1}^{3} \boldsymbol{r}_k \times \boldsymbol{r}_k^*$ is the world-frame small-angle orientation error, $\boldsymbol{p}_b, R_b$ the base pose, $\boldsymbol{\theta}$ the joint angles, $\boldsymbol{\gamma}$ the gimbal angles, and $\beta_{h,t}$ the toe-brace schedule.

Term by term

- **Swing pad.** For every pad that is released and not yet raking, the pad follows the Bezier approach arc $\boldsymbol{p}_h^*(t)$ toward its next anchor. W_h^p is a 3×3 diagonal weight; w_h^r ramps with arc progress (weak at the start of a hand swing, strong toward landing). This is an intent, not a fact: the arc encodes a preference (clear the wall, never drag unloaded spine tips across it), not a physical constraint.
- **Toe brace.** $W^{\text{toe}} = \text{diag}(300, 30, 30)$: only the wall-normal x component is enforced strongly; y and z ride freely, because the 2-DoF ankle cannot place the toe tip exactly in 3D.

- **Base intent.** $\boldsymbol{p}_b^*(t)$ blends linearly from the segment-start position to "press 3 cm toward the wall, shift 2.5 cm toward the supporting side, rise half a step"; $R_b^*(t)$ interpolates back to upright. The weights are small (30–100 for position, 2–50 for orientation), so the welds decide where the base can actually go.
- **Posture reference.** $\boldsymbol{\theta}^*(t)$ blends from the segment-start angles to the stance angles with the smallest weight (0.3); it only keeps directions that no other term constrains from wandering.
- **Limit-margin spring.** $W_{\text{lim}}(\boldsymbol{\theta})$ is nonzero only for joints inside the band within $\mu_s = 10\%$ of their range from either stop; it pulls such a joint back to the band boundary $\boldsymbol{\theta}_{\text{bnd}}$. It differs from the hard margin $\boldsymbol{m}$ of constraint (C4): the hard margin is a wall, the spring is an early nudge.
- **Gimbal return.** w_γ makes the solver prefer the arm joints over the passive gimbal when satisfying a weld; the gimbal turns only when the arm cannot.
- **Smoothness.** w_a in (1) penalizes the discrete acceleration; constraint (C7) bounds it, w_a makes the trajectory smoother than the bound alone requires.

Default weights

Weight	Value	Role
W_h^p (foot)	$\text{diag}(1000, 1000, 1000)$	swing-foot position
w_h^r (foot)	100	swing-foot orientation
W_h^p (hand)	$500 \to 2000$ with arc progress	swing-hand position
w_h^r (hand)	$50 \to 500$ with arc progress	swing-hand orientation
W^{toe}	$\text{diag}(300, 30, 30)$	toe brace
W^b, w_r^b	$\text{diag}(30 \dots 100)$, $2 \dots 50$	base intent
W_θ	0.3 (right-hand swing 10)	posture reference
W_{lim}, μ_s	3, 0.10	limit-margin spring
w_γ	3	gimbal return
w_a	to be tuned (order 1, in rad/s^2)	smoothness

The swing-hand weights follow $w = w_{\min} + (w_{\max} - w_{\min}) \max\big(1 - \min(s_a/0.3, 1),\ s_a^2\big)$ with arc progress s_a: strong at the segment start, weak mid-swing, strong again before landing, which spreads the re-equilibration caused by releasing a weld over the whole swing instead of a single frame.